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13 July 1955
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MILITARY SPECIFICATION

RADAR BOMBING EQUIPMENT AN/APA-16,

BENCH, PREFLIGHT, AND FLIGHT TEST, INSTRUCTIONS FOR

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 Scope: This specification establishes bench, preflight, and flight test instructions for Model AN/APA-16 Radar Bombing Equipment, a device for performing low altitude bombing and when used with TD-6/APA-16 Adapter Unit, for performing rocket firing.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and publications of the issue in effect on the date of invitation for bids forms a part of this specification.

SPECIFICATIONS:

Military

MIL-R-18895(Aer)	Radar Bombing Equipment, AN/APA-16
MIL-R-18983(Aer)	Radar Bombing Equipment, AN/APA-16; Installation Instructions for

PUBLICATIONS:

Handbook of Operation Instructions NAVAER 16-30APA16-505
Handbook of Service Instructions NAVAER 16-30APA16-506

(When requesting specifications, standards, drawings and publications refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania, Attention Technical Records Division.)

3. REQUIREMENTS

3.1 Reporting Procedure: All equipments are assumed to be in proper operating condition. The procedures prescribed herein define the methods for which the subject equipment should be tested before and after installation in the applicable aircraft. In the event the subject equipment does not satisfactorily pass the tests outlined herein, report of the condition shall be submitted to the local BAR or InsMat for subsequent transmittal to the Bureau of Aeronautics. All reports of unsatisfactory, defective material shall be made in accordance with existing directives.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests: The testing of the equipment shall be classified as follows:

a. Bench Tests: Bench tests are those tests performed prior to installation and shall be conducted at the aircraft contractor's plant to determine that the equipment has not been damaged and that the performance and operation has not been effected by shipping or handling. Contractors may request permission from the Bureau of Aeronautics to omit the bench tests if they can show that the test is not necessary in complying with the other requirements of this specification and/or the contract. This permission, if granted, may be revoked at any time at the discretion of the Bureau of Aeronautics. Notice of revocation will be in writing and shall be effective upon receipt by the contractor.

b. Preflight Tests: Preflight tests shall be conducted on each aircraft after the equipment has been installed in the aircraft and while the aircraft is still on the ground to determine that the equipment is installed properly to insure satisfactory operation.

c. Flight Tests: Flight tests shall be conducted on selected aircraft to insure satisfactory performance and operation of the equipment while the aircraft is in flight. This test is conducted to simulate ultimate operational environment, and as a check on vibration and electrical interference.

4.1.1 Determination of System Delay: Because of inherent time delays in the radar system, the AN/APA-16 and the airplane bomb release circuit, true range zero does not correspond to the radar trigger pulse. It is necessary to correct for this system delay and this is done with the aid of the Bomb Correction Dial since this dial is directly calibrated in feet of release range. The system delay shall be obtained in the manner described in Paragraphs 4.6.3 and 4.6.4, and this number placed in the space provided for it on the plate below the Range Zero Knob on the Control Unit. Since this delay is a function of the entire system from radar to bomb release shackles, it will vary with any change in the system and must be redetermined in case of such change.

WARNING: Operation of this equipment involves the use of high voltages which are dangerous to life. Personnel must observe safety precautions at all times.

4.2 Test Equipment: The following equipment, or equipment of equal or superior characteristics, is recommended for use in the performance of the tests required by this specification. Items which are considered peculiar to service use and which will be loaned by the Navy are marked with an asterisk (*). Other items may be supplied in special circumstances, but in all cases such circumstances must be justified by the contractor and adjudged by the Navy.

NAME	NOMENCLATURE	ASO STOCK NO.
* Range Calibrator	TS-102A/AP	R-16-AN-TS-102A/AP
Oscilloscope	TS-239/UP	R-16-AN-TS-239/UP
or		
Oscilloscope	TS-34/AP	R-16-AN-TS-34/AP
Multimeter (20,000 Ω/Volt)	TS-352/U	R-16-AN-TS-352/U
Tube Tester	TV-2/U	R-16-AN-TV-2/U

4.3 Bench Tests:

4.3.1 General: A permanently located bench or test rack for mounting the equipment should be provided. The rack should have cables available for simulating a normal installation in an aircraft and shall have a source of power available similar to that supplied by the aircraft. The power source should be capable of delivering two amperes at 115 volts and 400 to 1800 cycles per second, and approximately 0.5 amperes at 26.5 volts DC (plus 0.5 volt).

4.3.1.1 Test Procedure:

4.3.1.2 Visual Inspection: The equipment should be visually inspected for any damage due to shipping and handling.

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4.3.2 Mechanical Checks:

a. Check all connector lock rings for tightness.

b. Check plugs on coaxial cables for tightness.

c. Check all tubes for proper seating.

d. Check all units for obvious visible mechanical and electrical discrepancies.

e. Pull all knobs gently to check security.

f. Check presence and proper rating of all fuses.

g. Remove dust cover of Range Marker Unit and make certain that Switch S-201 is in position 2. Make sure that 6J6 tube is in V228 position and not in V218 position. (Refer to Paragraph 2-80 of Handbook of Service Instructions, NAVAER 16-30APA16-506).

h. Adjust the Range Mark Knob on the Control Unit so that it is free of the bracket in which it may be locked. (Refer to Paragraph 2-110 of Handbook of Service Instructions, NAVAER 16-30APA16-506).

i. Check that dials operate without binding or excessive looseness.

j. Check that terminals "C" and "D" on the release cable receptacle are strapped.

4.3.3 Electrical Checks, General:

a. Turn power on and let the equipment warm up for a few minutes.

b. Measure AC input voltage, current and frequency. (Standard voltage to be 115 volts plus or minus 1 volt).

4.3.3.1 Indicator Display Check: Check that the correct indicator display is presented, depending on the type of radar used. (Refer to Handbook of Operational Instructions NAVAER 16-30APA16-505, Figures 2-2, 2-3, 2-4, 2-5, and 2-6.)

4.4 *Operational Tests:*

4.4.1 Tests without a Radar: (Refer to Paragraph 4-22 of Handbook of Service Instructions, NAVAER 16-30APA16-506). If a radar system is not available, tests may be made with Range Calibrator TS-102A/AP to provide trigger pulses, and Oscilloscope TS-239/UP or TS-34/AP. (See Figure 4-4 of above referenced Handbook for interconnection diagram). The Range Marker Unit, the Control Unit and the Capacitor Unit are connected for normal use. Turn the Control Unit power on as follows:

Calibrator Settings:

Control or Switch	Position
MARKER AMPL	Approximately midway of its range
MARKER POLARITY	/
SYNC POLARITY	/
SYNC TERM	OFF
REPETITION FREQ	Set at repetition rate of associated radar or at 800
POWER	ON

4.4.2 Oscilloscope Settings: When using the Oscilloscope TS-34/AP, it will be necessary to connect a 150-OHM termination to ground at the SIGNAL INPUT jack of the Oscilloscope.

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Control or Switch	Position of Oscilloscope TS-34/AP
POWER	ON
SWEEP TIME or SPEED	MED
ADJ SWEEP TIME	---
SYNC POLARITY	POS
SWEEP SELECTOR	START-STOP
SYNC	EXT SYNC
INPUT IMPEDANCE	HIGH-40 DB
IMAGE SIZE	Adjust to upper reference lines

4.4.3 Release Range Calibration: (Refer to Paragraph 4-23 of Handbook of Service Instructions, NAVAER 16-30APA16-506). The step-by-step procedure outlined in the following paragraphs should be used for making release range and associated calibrations. As the serviceman gains experience, the procedure may be somewhat simplified.

4.4.4 Reference Range Zero: The purpose of this adjustment is to establish a reference point at a fixed and accurately reproducible time following the trigger pulse, from which to start the calibration procedure. Set up the controls in the Control Unit as follows:

Control	Position
POSITION	Nearly fully clockwise
BOMB SCALE ADJ	Midway of its range
CALIBRATE SWITCH	REF RANGE ZERO
BOMB CORRECTION	Zero feet
RANGE ZERO	Fully counterclockwise
SEARCH-TRACK	SEARCH

a. Switch to TRACK and turn the RANGE ZERO control slowly clockwise until the dial lights go out.

b. Check this setting several times by turning the BOMB CORRECTION dial toward -100, switching to TRACK, and then advancing the BOMB CORRECTION dial slowly until the lights go out again.

c. The BOMB CORRECTION dial should read 0 ± 5 feet. If this tolerance is exceeded, make minor corrections of the RANGE ZERO control and repeat the above check. When adjusted, tighten the knurled nut under the RANGE ZERO control knob, and recheck to be sure that tightening the nut has not disturbed the RANGE ZERO control setting.

d. The purpose of this adjustment is to set one of the marker pulses from the calibrator at a definite and reproducible point with respect to the reference range zero established in the preceding adjustment, so that calibrating markers are accurately located 1,500 feet, 3,000 feet, etc., in range from that range zero point. Set up the controls in the Control Unit as follows:

Control	Position
CALIBRATE SWITCH	RATE OR RLS RANGE ZERO
BOMB CORRECTION	∠ 50 feet

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e. Switch to TRACK and turn the PHASE ADJ control of the calibrator slowly in a counterclockwise direction until the dial lights go out.

f. Check this setting several times, each time turning the BOMB CORRECTION dial toward -100, switching to TRACK, and then advancing the BOMB CORRECTION dial slowly toward ∠ 50 until the dial lights again go out.

g. The dial reading should be ∠ 50 ± 5 feet. If the reading exceeds this tolerance, turn the PHASE ADJ control slightly clockwise, set the BOMB CORRECTION dial to ∠ 50, and repeat the two previous steps. Repeat again if necessary to obtain the proper setting.

4.4.5 Calibrating the BOMB CORRECTION Scale: The purpose of this calibration is to adjust the voltage across the BOMB CORRECTION potentiometer to a value such that a given change in feet on the BOMB CORRECTION dial will produce a corresponding change in the release range. As a preliminary step, it is necessary to adjust the computer in the Control Unit so that, for the setting of the CLOSING RATE and ALTITUDE Dials specified, the release range pulse occurs at the correct distance from range zero. Set up the controls in the Control Unit as follows:

Control	Position
CLOSING RATE	325 Knots
BOMB CORRECTION	∠ 50 feet
ALTITUDE	481 feet
CALIBRATE SWITCH	RELEASE RANGE ADJ
RLS RANGE ADJ	Maximum Counterclockwise

a. Switch to TRACK and turn the RLS RANGE ADJ control slowly in a clockwise direction until the dial lights go out. Check the setting with the BOMB CORRECTION dial as in previous tests. Lights should go out at ∠ 50 ± 10 feet.

b. Retard the BOMB CORRECTION dial toward -100, turn the CALIBRATE SWITCH to RATE OR RLS RANGE ZERO, and check range zero with the BOMB CORRECTION dial. The reading should be ∠ 50 ± 10 feet. If exceeded, set the BOMB CORRECTION dial on ∠ 50 feet and readjust the PHASE ADJUST control of the calibrator, as in 4.4.4.(a).

c. Return the CALIBRATE SWITCH to RELEASE RANGE ADJ. Check the BOMB CORRECTION dial reading as before, and readjust the RLS RANGE ADJ control. Continue steps 4.4.4.(f) and 4.4.4.(g) until the dial lights go out on ∠ 50 ± 10 feet on both positions of the CALIBRATE SWITCH.

d. Move the ALTITUDE dial to read 348 feet. Switch to TRACK, and advance the BOMB CORRECTION dial slowly toward ∠ 500; the dial lights should go out when ∠ 500 is reached. If they do not, adjust the BOMB SCALE ADJ control until the light will go out at ∠ 500 ± 5 feet.

e. Repeat checks of the settings in the previous four steps, adjusting respective controls if necessary since the RANGE ZERO and BOMB SCALE ADJ settings are reached by a process of successive approximations. When properly adjusted, the spread between readings on the BOMB CORRECTION dial for ALTITUDE dial settings of 481 and 348 feet should be 450 ± 20 feet.

4.4.6 Rate Zero Check: The purpose of this check is to adjust the rate sweep generator in the Range Marker Unit for constant output voltage when zero rate voltage is applied to it. Set up the controls of the voltage unit and test equipment as follows:

Control	Location	Position
POSITION	Control Unit	Near fully clockwise
ALTITUDE	Control Unit	50 feet
BOMB CORRECTION	Control Unit	-100 feet
CALIBRATE SWITCH	Control Unit	OPERATE
CLOSING RATE	Control Unit	300 knots

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Control	Location	Position
RANGE MARK	Control Unit	Adjust as required
MARKER POLARITY	Calibrator	- (Minus)

a. Adjust the test oscilloscope to obtain clearcut images. If oscilloscope CW-60ACB is used, the image amplitude should be adjusted to one-half inch or less to prevent overloading.

b. While observing the test oscilloscope screen, turn the SEARCH-TRACK switch on the Control Unit to TRACK.

c. The range and release markers should appear on the sweep trace as narrow vertical pulses. The calibrating pulses will appear as equispaced bright spots along the sweep trace.

d. Use the POSITION control to set the tracking edge of the range-mark image within about 1/16 inch of the nearest calibrating marker.

e. Observe the screen for a period of 1 minute. If there is any relative motion between the range mark and the calibrating marker, adjust the RATE ZERO control to stop such motion. After adjustment, check the screen for another minute period. Repeat the above procedure if necessary until no motion is apparent for a 1 minute period.

4.4.7 Interconnection of C-275A/APA-16 and SN-12B/APA-16 with TD-6/APA-16: (Refer to the Difference Data Sheet "Model AN/APA-16 Aircraft Radar Equipment Modified for use with Adapter Unit TD-6/APA-16"). Whenever the AN/APA-16 is utilized for firing rockets, modifications to the interconnecting cables, as well as the required tests and calibration procedures relating to TD-6/APA-16, shall be performed.

4.5 Preflight Tests

4.5.1 General: After equipment has been installed in the aircraft, the following checks shall be made using an external power supply.

4.5.1.1 Test Procedure

4.5.1.2 Input AC power supply shall be 115 volts, plus or minus 5 volts. This check should be made with all electronic equipment in the airplane turned on and with engine driven generators operating through variable speeds within safe limits.

a. AN/APA-16 range mark is visible on radar scope.

b. Range mark moves down scope when equipment is on "Track".

c. Proper operation of all Control Unit knobs, switches, lights and dials.

d. Now repeat the test procedure outlined above from 4.4.1 through 4.4.6.

e. Assure that bomb release signal is furnished to armament bomb circuit and that bomb shackles in the airplane operate properly when energized by the AN/APA-16 release signal.

f. BOMB CORRECTION dial is set at plus 300 and RANGE ZERO knob (on REF RANGE ZERO) set for release at that range. BOMB CORRECTION dial shall then be returned to zero. This puts the equipment in readiness for bombing.

4.6 Flight Tests

4.6.1 General: Flight tests shall be conducted on the first three aircraft of the contract and on every tenth aircraft thereafter. (Except that, at the discretion of the Bureau of Aeronautics Representative, not more than one aircraft per month need be flight tested) unless directed otherwise by the procuring agency. Acceptance of an aircraft radar equipment installation shall be based upon installation procedures which will permit the attainment of the bombing performance requirements contained in the following test procedure. The above test procedure (Sections 4.3 and 4.5)

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will normally insure satisfactory operation. The final criterion, however, is the flight test, and no installation will be considered to comply with this specification unless the performance specified in this section is, or can be, obtained.

4.6.1.1 Test Procedure: Repeat the test procedure outlined above from 4.4.1 through 4.4.6 and then continue with the following tests:

4.6.2 Rate Calibration: The purpose of this calibration is to check the rate linearity and adjust the rate voltage from the computer so that it will correctly control the rate sweep generator in accordance with calibrations on the CLOSING RATE dial. Set the controls on the AN/APA-16 Control Unit and test equipment as follows:

Control	Location	Position
POSITION	Control Unit	Near fully clockwise
ALTITUDE	Control Unit	50 feet
BOMB CORRECTION	Control Unit	-100 feet
CALIBRATE SWITCH	Control Unit	OPERATE
CLOSING RATE	Control Unit	As specified in text below
RANGE MARK	Control Unit	Released
SWEEP SPEED	TS-34/AP	MED
SWEEP TIME	CW-60ACB	NORMAL
MARKER POLARITY	Calibrator	- (Minus)

4.6.2.1 Set the CLOSING RATE dial to 148 knots and, while observing the oscilloscope screen, switch to TRACK. The range mark should move to the left along the sweep trace. Beginning with the third marker to appear, carefully measure the time required to pass five additional markers. It should require 30 seconds. If it does not, correct the RATE ADJ control, turning it clockwise if the elapsed time is greater, and counterclockwise if less than it should be. Check the time required to uncover or pass other groups of five timing markers, resetting the SEARCH-TRACK switch if necessary. Make the final adjustment accurately on a group of five markers near the bottom of the screen, after switching to the shortest sweep. Do not use the last two timing markers to appear.

4.6.3 Dry Run: Dry bombing run flight tests shall normally be required only in aircraft tested in accordance with paragraph 4.6.3.6) below. While flying at an altitude of 350 to 500 feet pick out a target dead ahead about 5 miles. Have pilot fly toward it and note whether the range mark follows target properly while tracking. If satisfactory, proceed with system delay bombing tests.

4.6.4 Determination of System Delay Compensation: The preceding calibrations have all been made with respect to an arbitrary range zero. In actual operation the release points must be located with respect to true range zero which is dependent on the system delay. This is most conveniently done with the aid of the BOMB CORRECTION potentiometer by setting its dial at a point other than zero when making reference range-zero checks in flight. A space is provided below the RANGE ZERO control for entering the proper setting for this dial.

a. The system delay is usually not known and is difficult to measure directly. A practical solution is to make an approximate allowance for it and then check results by dropping practice bombs. A bombardier familiar with the equipment should operate it during these tests. The procedure is given below:

b. When the calibrations specified in previous paragraphs have been completed on the ground, proceed to make flight tests, operating the equipment as instructed in the Handbook of Operation Instructions NAVAER 16-30APA16-505.

c. Set the BOMB CORRECTION dial at ∔ 300 feet when making the reference range zero adjustment. This adjustment should be checked prior to each of these test bombing runs.

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d. After making a reference range zero check, be sure to return the BOMB CORRECTION dial to zero or to the correct value to allow for bomb "trail". If such an allowance is to be made, set in the correct number of feet in a direction toward -100 on the BOMB CORRECTION dial.

e. Make bombing runs at an altitude of 350 feet on a small target of known size, dropping a number of practice bombs (not less than 12 are suggested). Determine the point of impact of each with respect to the target, recording distance from target and location with respect to the course of the aircraft.

f. Determine the range error (eliminate azimuth error, which is not due to the AN/APA-16 equipment), assigning "+" signs to those cases in which the bomb overshoots the target and "-" signs to the cases where the bomb falls short of the target.

g. On a vertical line select a point representing the center of the target and plot errors in range to scale "+" values above and "-" values below the target location. Then find the center of the smallest circle that will enclose 80% of the points. Determine the distance from the center of this circle to the point representing the center of the target, assigning the proper sign to it.

h. If this distance is negative, add it to 300 feet; if positive, subtract it from 300 feet. Enter the result obtained in the space provided on the plate below the RANGE ZERO knob on the Control Unit.

i. This value should be set thereafter on the BOMB CORRECTION dial when adjusting the checking reference range zero during flight. No change should be made in the value unless careful observations show that bombs are consistently falling short or overshooting.

j. A tabulation of the bombing results obtained on each flight test shall be recorded in a suitable form, including a plot, which indicates the error of individual drops, the type, number, and sequence of bombs dropped on each flight, the calculated MPI, and two (2) copies forwarded with applicable comments, if any, to the Bureau of Aeronautics.

4.7 Final Check - After the flight test a visual inspection shall be made to determine whether any ill effects from the flight are apparent. Particular attention shall be given to the supports and securing mechanisms.

5. PREPARATION FOR DELIVERY

5.1 Not Applicable

6. NOTES

6.1 Not Applicable

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